

Lab 03:

Constructor & Destructor

Question#01:

CODE:

```
CT-25276_Q1.cpp > Complex > Complex(const Complex &)\n1  #include <iostream>\n2  using namespace std;\n3\n4  class Complex {\n5      private:\n6          int* real;\n7          int* imag;\n8\n9      public:\n10         Complex(int r, int i) {\n11             real = new int;\n12             imag = new int;\n13\n14             *real = r;\n15             *imag = i;\n16         }\n17\n18         Complex(const Complex& obj) {\n19             real = new int;\n20             imag = new int;\n21\n22             *real = *(obj.real);\n23             *imag = *(obj.imag);\n24\n25\n26         }\n27\n28         void display() {\n29             cout << "real: " << *real << " imag: " << *imag << endl;\n30         }\n31\n32         ~Complex() {\n33             delete real;\n34             delete imag;\n35         }\n36     };\n37\n38     int main() {\n39\n40         Complex c1(10, 20);\n41         Complex c2 = c1;\n42\n43         c1.display();\n44         c2.display();\n45     }
```

OUTPUT:

```
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 03> cd "d:\NE
g++ CT-25276_Q1.cpp -o CT-25276_Q1 } ; if ($?) { .\CT-25276_Q1 }
real: 10 imag: 20
real: 10 imag: 20
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 03> |
```

QUESTION#02:

CODE:

```
CT-25276_Q2.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  class Character
5  {
6      // member variables
7      int *health, *power, *defense;
8  public:
9      // setter methods
10     void setHealth(int input)
11     {
12         *health = input;
13     }
14     void setPower(int input)
15     {
16         *power = input;
17     }
18     void setDefense(int input)
19     {
20         *defense = input;
21     }
22     // getter methods
23     int getHealth()
24     {
25         return *health;
26     }
27     int getPower()
28     {
29         return *power;
30     }
31     int getDefense()
32     {
33         return *defense;
34     }
35     // Constructors
36     Character()
37     {
38         health = new int(100);
39         power = new int(0);
40         defense = new int(0);
41     }
42     Character(int h, int p, int d)
43     {
```

```
43     {
44         health = new int(h);
45         power = new int(p);
46         defense = new int(d);
47     }
48     Character(const Character &obj)
49     {
50         health = new int(*(obj.health));
51         power = new int(*(obj.power));
52         defense = new int(*(obj.defense));
53     }
54 };
55
56 int main()
57 {
58     Character a, b(80, 200, 500);
59     Character c = a;
60
61     cout << "Character A:" << endl
62         << "Health: " << a.getHealth() << endl
63         << "Power: " << a.getPower() << endl
64         << "Defense: " << a.getDefense() << endl
65         << endl;
66
67     cout << "Character B:" << endl
68         << "Health: " << b.getHealth() << endl
69         << "Power: " << b.getPower() << endl
70         << "Defense: " << b.getDefense() << endl
71         << endl;
72
73     cout << "Character C:" << endl
74         << "Health: " << c.getHealth() << endl
75         << "Power: " << c.getPower() << endl
76         << "Defense: " << c.getDefense() << endl;
77
78     return 0;
79 }
80
```

OUTPUT:

```
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals> g++ CT-25276_Q2.cpp -o CT-25276_Q2 } ; if ($?) { .\CT-25276_Q2 }
Character A:
Health: 100
Power: 0
Defense: 0

Character B:
Health: 80
Power: 200
Defense: 500

Character C:
Health: 100
Power: 0
Defense: 0
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals>
```

QUESTION#03:

CODE:

Object Oriented Programming C++

```
CT-25276_Q3.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class TollBooth
5  {
6      int totalCars;
7      double totalMoney;
8  public:
9
10     void payingCar(void)
11     {
12         totalCars++;
13         totalMoney += 0.50;
14     }
15     void displayTollData()
16     {
17         cout << "Total Cars: " << totalCars << endl
18             << "Total Money: $" << totalMoney << endl
19             << endl;
20     }
21
22     TollBooth() : totalCars(0), totalMoney(0.0) {}
23 };
24
25 int main()
26 {
27
28     TollBooth hyderabadStation;
29
30     cout << "Morning Shift:-" << endl;
31     for (int i = 0; i < 50; i++)
32         hyderabadStation.payingCar();
33     hyderabadStation.displayTollData();
34
35     cout << "Evening Shift:-" << endl;
36     for (int i = 0; i < 100; i++)
37         hyderabadStation.payingCar();
38     hyderabadStation.displayTollData();
39
40     cout << "Night Shift:-" << endl;
41     for (int i = 0; i < 10; i++)
42         hyderabadStation.payingCar();
43     hyderabadStation.displayTollData();
44     return 0;
45 }
```

OUTPUT:

```
PS D:\NED University\2nd Semester\Object Ori  
(OOPs)\Practicals\Lab 03\" ; if ($?) { g++ C  
Morning Shift:-  
Total Cars: 50  
Total Money: $25  
  
Evening Shift:-  
Total Cars: 150  
Total Money: $75  
  
Night Shift:-  
Total Cars: 160  
Total Money: $80  
  
PS D:\NED University\2nd Semester\Object Ori
```

QUESTION#04:

CODE:

```
CT-25276_Q4.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  class BookType
5  {
6      string title, author[4], pub;
7      int ISBN, noOfCpy, noOfAuth;
8      float price;
9
10 public:
11     // operations on title
12     string getTitle()
13     {
14         return title;
15     }
16     void setTitle(const string t)
17     {
18         title = t;
19     }
20     bool checkTitle(const string t)
21     {
22         return title == t;
23     }
24     // operations of number of copies in stock
25     int getCopies()
26     {
27         return noOfCpy;
28     }
29     void setCopies(const int num)
30     {
31         noOfCpy = num;
32     }
33     void updateCopies(const int num)
34     {
35         noOfCpy += num;
36         if (noOfCpy < 0)
37             noOfCpy = 0;
38     }
```

Object Oriented Programming C++

```
39 // operations on authors
40 void setAuth(const string auth)
41 {
42     if (noOfAuth < 4)
43         author[noOfAuth++] = auth;
44     else
45         cout << "Max Limit Reached!!\n";
46 }
47 void getAuth()
48 {
49     cout << "No. of authors: " << noOfAuth << endl;
50     for (int i = 0; i < noOfAuth; i++)
51     {
52         cout << "\tAuthor" << i + 1 << ": " << author[i] << endl;
53     }
54 }
55 // operations on Publisher
56 string getPub()
57 {
58     return pub;
59 }
60 void setPub(const string t)
61 {
62     pub = t;
63 }
64 // operations on book price
65 float getPrice()
66 {
67     return price;
68 }
69 void setPrice(float p)
70 {
71     price = p;
72 }
73 // operations on ISBN
74 int getISBN()
75 {
76     return ISBN;
77 }
78 void setISBN(int i)
79 {
80     ISBN = i;
81 }
```

Object Oriented Programming C++

```
82     // constructor
83     BookType()
84     {
85         pub = title = "Unknown";
86         for (int i = 0; i < 4; i++)
87             author[i] = "N/A";
88         ISBN = noOfCpy = noOfAuth = price = 0;
89     }
90 };
91
92 void listFeatures(BookType book)
93 {
94     cout << "Title: " << book.getTitle() << endl;
95     cout << "Publisher: " << book.getPub() << endl;
96     cout << "Price: $" << book.getPrice() << endl;
97     cout << "ISBN: " << book.getISBN() << endl;
98     book.getAuth();
99     int copies = book.getCopies();
100     if (copies == 0)
101         cout << "\033[1;31mUnavailable.\033[0m" << endl;
102     else
103         cout << "\033[1;32mIn Stock " << copies << " copies available.\033[0m" << endl;
104 }
105
```

Object Oriented Programming C++

```
105
106 int main()
107 {
108     BookType books[100];
109     cout << "====Welcome to ABC Book Shop!====" << endl;
110     int choice, totalBooks = 0;
111     do
112     {
113         cout << "\nSelect options from below:" << endl;
114         cout << "1. Add a book" << endl;
115         cout << "2. Display all books" << endl;
116         cout << "3. Search by Title" << endl;
117         cout << "4. Search by ISBN" << endl;
118         cout << "5. Search by Publisher" << endl;
119         cout << "6. Update number of copies" << endl;
120         cout << "7. Exit" << endl;
121         cin >> choice;
122         cout << endl;
123         switch (choice)
124         {
125             case 1:
126                 if (totalBooks < 100)
127                 {
128                     string title, author, pub;
129                     int ISBN, noOfCpy, noOfAuth;
130                     float price;
131                     cout << "Enter title of the book: ";
132                     cin >> title;
133                     books[totalBooks].setTitle(title);
134                     cout << "Enter number of authors: ";
135                     cin >> noOfAuth;
136                     if (noOfAuth > 4)
137                         cout << "Only 4 authors allowed at max.\n";
138                     for (int i = 0; i < noOfAuth && i < 4; i++)
139                     {
140                         cout << "Enter the name of author " << i + 1 << ": ";
141                         cin >> author;
142                         books[totalBooks].setAuth(author);
143                     }
144                     cout << "Enter the name of publisher: ";
145                     cin >> pub;
146                     books[totalBooks].setPub(pub);
147                     cout << "Enter the ISBN number: ";
148                     cin >> ISBN;
149                     books[totalBooks].setISBN(ISBN);
150                     cout << "Enter the number of copies available in stock: ";
```

Object Oriented Programming C++

```
150         cout << "Enter the number of copies available in stock: ";
151         cin >> noOfCpy;
152         books[totalBooks].setCopies(noOfCpy);
153         cout << "Enter the price of the book: $";
154         cin >> price;
155         books[totalBooks].setPrice(price);
156         totalBooks++;
157     }
158     else
159         cout << "Max limit of books entered." << endl;
160     break;
161 case 2:
162     cout << "\033[1;33mTotal number of books (" << totalBooks << "):\033[0m" << endl;
163     for (int i = 0; i < totalBooks; i++)
164     {
165         void listFeatures(BookType book) =====\033[0m" << endl;
166         listFeatures(books[i]);
167         cout << "\033[1;34m===== \033[0m" << endl;
168     }
169     break;
170 case 3:
171 {
172     string title;
173     cout << "Enter the title: ";
174     cin >> title;
175     int matches = 0;
176     for (int i = 0; i < totalBooks; i++)
177         if (books[i].checkTitle(title))
178         {
179             matches++;
180             cout << "\033[1;34m===== \033[0m" << endl;
181             listFeatures(books[i]);
182             cout << "\033[1;34m===== \033[0m" << endl;
183         }
184     cout << "\033[1;33mFound " << matches << " matches.\033[0m" << endl;
185 }
186 break;
187 case 4:
188 {
189     int ISBN;
190     cout << "Enter the ISBN: ";
191     cin >> ISBN;
```

Object Oriented Programming C++

```
191     cin >> ISBN;
192     bool found = false;
193     for (int i = 0; i < totalBooks; i++)
194     {
195         if (books[i].getISBN() == ISBN)
196         {
197             cout << "\033[1;34m===== \033[0m" << endl;
198             listFeatures(books[i]);
199             cout << "\033[1;34m===== \033[0m" << endl;
200             found = true;
201             break;
202         }
203     }
204     if (!found)
205     {
206         cout << "\033[1;31mThe book with ISBN " << ISBN << " doesn't exists. \033[0m" << endl;
207     }
208     break;
209     case 5:
210     {
211         string pub;
212         cout << "Enter the Publisher: ";
213         cin >> pub;
214         int matches = 0;
215         for (int i = 0; i < totalBooks; i++)
216         {
217             if (books[i].getPub() == pub)
218             {
219                 matches++;
220                 cout << "\033[1;34m===== \033[0m" << endl;
221                 listFeatures(books[i]);
222                 cout << "\033[1;34m===== \033[0m" << endl;
223             }
224         }
225         cout << "\033[1;33mFound " << matches << " matches. \033[0m" << endl;
226     }
227     break;
228     case 6:
229     {
230         int ISBN, i;
231         cout << "Enter the ISBN: ";
232         cin >> ISBN;
233         bool found = false;
234         for (i = 0; i < totalBooks; i++)
235         {
236             if (books[i].getISBN() == ISBN)
237             {
```

Object Oriented Programming C++

```
231         {
232             cout << "\033[1;34m===== \033[0m" << endl;
233             listFeatures(books[i]);
234             cout << "\033[1;34m===== \033[0m" << endl;
235             found = true;
236             break;
237         }
238     if (!found)
239         cout << "\033[1;31mThe book with ISBN " << ISBN << " doesn't exists. \033[0m" << endl;
240     else {
241         int cpy;
242         cout << "Current number of copies: " << books[i].getCopies() << endl;
243         cout << "Add number of copies: ";
244         cin >> cpy;
245         books[i].updateCopies(cpy);
246     }
247 }
248 break;
249 case 7:
250     cout << "Exiting..." << endl;
251     break;
252 default:
253     cout << "\033[1;31mInvalid option \033[0m" << endl;
254     break;
255 }
256 } while (choice != 7);
257 return 0;
258 }
259 |
```

OUTPUT:

```
25276_Q4 } ; if ($?) { .\CT-25276_Q4  
}
```

====Welcome to ABC Book Shop!====

Select options from below:

1. Add a book
 2. Display all books
 3. Search by Title
 4. Search by ISBN
 5. Search by Publisher
 6. Update number of copies
 7. Exit
- 1

Enter title of the book: abc

Enter number of authors: 2

Enter the name of author 1: def

Enter the name of author 2: ghi

Enter the name of publisher: kim

Enter the ISBN number: 1234

Enter the number of copies available
in stock: 24

Enter the price of the book: \$9

Select options from below:

1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit

1

Enter title of the book: lmno

Enter number of authors: 5

Only 4 authors allowed at max.

Enter the name of author 1: ijk

Enter the name of author 2: kji

Enter the name of author 3: opq

Enter the name of author 4: qpo

Enter the name of publisher: ghi

Enter the ISBN number: 5678

Enter the number of copies available
in stock: 52

Enter the price of the book: \$6

Select options from below:

1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit

2

2

Total number of books (2):

=====

Title: abc

Publisher: kim

Price: \$9

ISBN: 1234

No. of authors: 2

Author1: def

Author2: ghi

In Stock 24 copies available.

=====

=====

Title: lmno

Publisher: ghi

Price: \$6

ISBN: 5678

No. of authors: 4

Author1: ijk

Author2: kji

Author3: opq

Author4: qpo

In Stock 52 copies available.

=====

Select options from below:

1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit

3

```
Enter the title: abc
=====
Title: abc
Publisher: kim
Price: $9
ISBN: 1234
No. of authors: 2
    Author1: def
    Author2: ghi
In Stock 24 copies available.
=====
Found 1 matches.

Select options from below:
1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit
3

Enter the title: abcd
Found 0 matches.

Select options from below:
1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit
4
```

```
Enter the ISBN: 12345
The book with ISBN 12345 doesn't exist.
```

```
Select options from below:
```

1. Add a book
 2. Display all books
 3. Search by Title
 4. Search by ISBN
 5. Search by Publisher
 6. Update number of copies
 7. Exit
- 4

```
Enter the ISBN: 1234
```

```
=====
```

```
Title: abc
```

```
Publisher: kim
```

```
Price: $9
```

```
ISBN: 1234
```

```
No. of authors: 2
```

```
    Author1: def
```

```
    Author2: ghi
```

```
In Stock 24 copies available.
```

```
=====
```

```
Select options from below:
```

1. Add a book
 2. Display all books
 3. Search by Title
 4. Search by ISBN
 5. Search by Publisher
 6. Update number of copies
 7. Exit
- 5

Enter the Publisher: klm

Found 0 matches.

Select options from below:

1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit

6

Enter the ISBN: 1234

=====

Title: abc

Publisher: kim

Price: \$9

ISBN: 1234

No. of authors: 2

Author1: def

Author2: ghi

In Stock 24 copies available.

=====

Current number of copies: 24

Add number of copies: 6

Select options from below:

1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit

2

```
Total number of books (2):
=====
Title: abc
Publisher: kim
Price: $9
ISBN: 1234
No. of authors: 2
    Author1: def
    Author2: ghi
In Stock 30 copies available.
=====
=====
Title: lmno
Publisher: ghi
Price: $6
ISBN: 5678
No. of authors: 4
    Author1: ijk
    Author2: kji
    Author3: opq
    Author4: qpo
In Stock 52 copies available.
=====

Select options from below:
1. Add a book
2. Display all books
3. Search by Title
4. Search by ISBN
5. Search by Publisher
6. Update number of copies
7. Exit
8
```

Object Oriented Programming C++

Invalid option

Select options from below:

1. Add a book
 2. Display all books
 3. Search by Title
 4. Search by ISBN
 5. Search by Publisher
 6. Update number of copies
 7. Exit
- 7

Exiting...

PS D:\NED Univ> .\ity\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 03>